10.2.1. History

Multimeter was invented by a British telecom engineer Donald Macadie. He had the difficulty of carrying many separate instruments while troubleshooting the telecommunication circuits. To avoid these difficulties, he invented an instrument which could measure Ampere, Volt and Ohm, which was named as AVO meter. Earlier, we used separate meters to measure multi-ranges of AC and DC voltages, AC and DC currents and resistance values. Figure 10.1 shows the separate range AC voltmeters.



Figure 10.2 Analog Multimeter Front Panel with Dial and Selector Switch

For example, to measure 230 volts AC, we used a voltmeter of 0-300 V range, whereas for measuring 6 V AC a 0-10 V AC voltmeter was recommended to measure the voltage accurately.

Using a single meter, we can measure AC and DC voltages (various ranges such as 0-10 V, 0-250 V), AC and DC currents (various ranges such as 0-0.25 A, 0-10 A) and resistance values (various ranges x 1, x 10, x 1 kΩ etc.). So it is called as multimeter or multi-tester or AVO meter. Using a multimeter, the measurements carried out using 20 separate meters can be measured in a single multimeter as shown in Figure 10.2.

To understand the various functions available in the multimeter, you should keenly watch the front panel of a multimeter. In a Multimeter, you can see 20 switching positions indicating various ranges of AC and DC voltage, AC and DC current and resistance values as shown in Figure 10.3. Thus, we can conclude that a single multimeter can be used in place of 20 separate meters.



Figure 10.3 Analog Multimeter Front Panel Selector Switch (Enlarged View)

ELEC_EQUIP_Chapter10.indd 203 11/10/2020 11:55:36 AM